

BLUESTOCK MUTUAL FUND ANALYTICS CAPSTONE PROJECT

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Project Title:

Mutual Fund Analytics Platform – End-to-End Data Engineering, Analytics, and Dashboarding Solution

1. EXECUTIVE SUMMARY

The mutual fund industry has experienced significant growth over the last decade due to increasing financial awareness, digital investment platforms, and systematic investment plans (SIPs). As the number of investors, fund houses, and schemes continues to grow, the volume of mutual fund data generated daily has become enormous. This creates a need for analytical systems capable of transforming raw financial data into actionable business insights.

The objective of this project was to build a comprehensive Mutual Fund Analytics Platform capable of collecting, cleaning, analyzing, and visualizing mutual fund data to support investment decision-making and business intelligence activities. The project integrates data engineering, exploratory data analysis, financial performance analysis, investor behavior analysis, and dashboard development into a single end-to-end analytics pipeline.

The project utilized multiple datasets including mutual fund master data, NAV history, AUM statistics, SIP inflows, category inflows, investor transactions, portfolio holdings, and benchmark indices. Data was processed through an ETL pipeline where raw information was cleaned, standardized, validated, and transformed into analysis-ready datasets.

The analytical phase focused on identifying performance trends, evaluating risk-return characteristics, studying investor behavior patterns, and measuring portfolio concentration. Advanced financial metrics such as Sharpe Ratio, Historical Value at Risk (VaR), Conditional Value at Risk (CVaR), and Herfindahl-Hirschman Index (HHI) were computed to provide deeper insights into fund performance and risk characteristics.

Interactive dashboards were developed in Tableau to provide stakeholders with a visual overview of industry trends, fund performance, investor analytics, and SIP market trends. These dashboards allow users to interact with data through filters and drill-down capabilities while exploring key investment insights.

The final outcome of the project is a complete mutual fund analytics ecosystem consisting of ETL pipelines, advanced analytics notebooks, risk analysis models, interactive dashboards, documentation, and presentation materials.

2. INTRODUCTION

The Indian mutual fund industry has witnessed rapid expansion driven by increasing retail participation, digital onboarding platforms, and rising awareness regarding long-term wealth creation. With thousands of mutual fund schemes and millions of investors actively participating in the market, there is a growing demand for data-driven investment analysis.

Investment decisions require understanding not only returns but also risk exposure, portfolio concentration, investor behavior, and market trends. Traditional reporting systems often provide limited visibility into these factors. Therefore, organizations require advanced analytics platforms that can transform large volumes of financial data into meaningful business intelligence.

This project addresses that requirement by developing an integrated analytics platform capable of:

- Monitoring mutual fund industry growth.
- Evaluating fund performance.
- Measuring investment risk.
- Understanding investor behavior.
- Tracking SIP trends.
- Supporting investment recommendations.

The platform combines data engineering, statistical analysis, and dashboard visualization to provide a complete analytical view of the mutual fund ecosystem.

3. PROJECT OBJECTIVES

The primary objectives of this project were:

1. Build a complete ETL pipeline for mutual fund datasets.
2. Clean and standardize raw financial data.
3. Perform exploratory data analysis on industry trends.
4. Evaluate mutual fund performance using financial metrics.
5. Analyze investor behavior and transaction patterns.
6. Measure risk using VaR and CVaR.
7. Develop interactive business intelligence dashboards.

8. Generate actionable business recommendations.

4. DATA SOURCES

The project utilized multiple datasets covering different aspects of the mutual fund ecosystem.

Dataset	Purpose
Fund Master	Scheme metadata and classification
NAV History	Historical net asset values
AUM by Fund House	Asset under management analysis
Monthly SIP Inflows	SIP trend analysis
Category Inflows	Category-level investment flows
Industry Folio Count	Investor participation analysis
Scheme Performance	Returns, Sharpe Ratio, Risk Metrics
Investor Transactions	Investor behavior analysis
Portfolio Holdings	Portfolio concentration analysis
Benchmark Indices	Market comparison and benchmarking

Raw Data Sources

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Fund Master

NAV History

AUM Data

Transactions

Portfolio Holdings

Benchmark Data

↓

ETL Pipeline

↓

Analytics Layer

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Dashboard Layer

5. ETL DESIGN AND DATA ENGINEERING

The ETL (Extract, Transform, Load) process forms the backbone of the analytics platform.

Extraction

Data was extracted from multiple CSV sources representing various aspects of the mutual fund ecosystem. These files contained historical and current information regarding fund performance, investor activity, portfolio holdings, and industry statistics.

Transformation

Several data cleaning operations were performed:

- Removal of missing values.
- Standardization of date formats.
- Data type conversion.
- Duplicate record elimination.
- Column name normalization.
- Feature engineering.
- Risk metric calculation.

Examples of engineered features include:

- Daily Returns
- Rolling Sharpe Ratio
- Risk Grade
- Investor Cohorts
- SIP Continuity Metrics

Loading

Cleaned datasets were exported into processed CSV files and used for dashboard development and advanced analytics.

Raw CSV

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Cleaning

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Transformation

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Feature Engineering

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Analytics

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Dashboard

6. EXPLORATORY DATA ANALYSIS (EDA)

EDA was conducted to understand overall market trends and identify patterns before advanced analytics were performed.

Industry Growth

Analysis of industry-wide AUM revealed significant growth between 2022 and 2025. This indicates increasing investor participation and market confidence.

Key Observations

- Industry AUM consistently increased over time.
- Large fund houses dominated market share.
- Equity-oriented funds attracted significant investments.

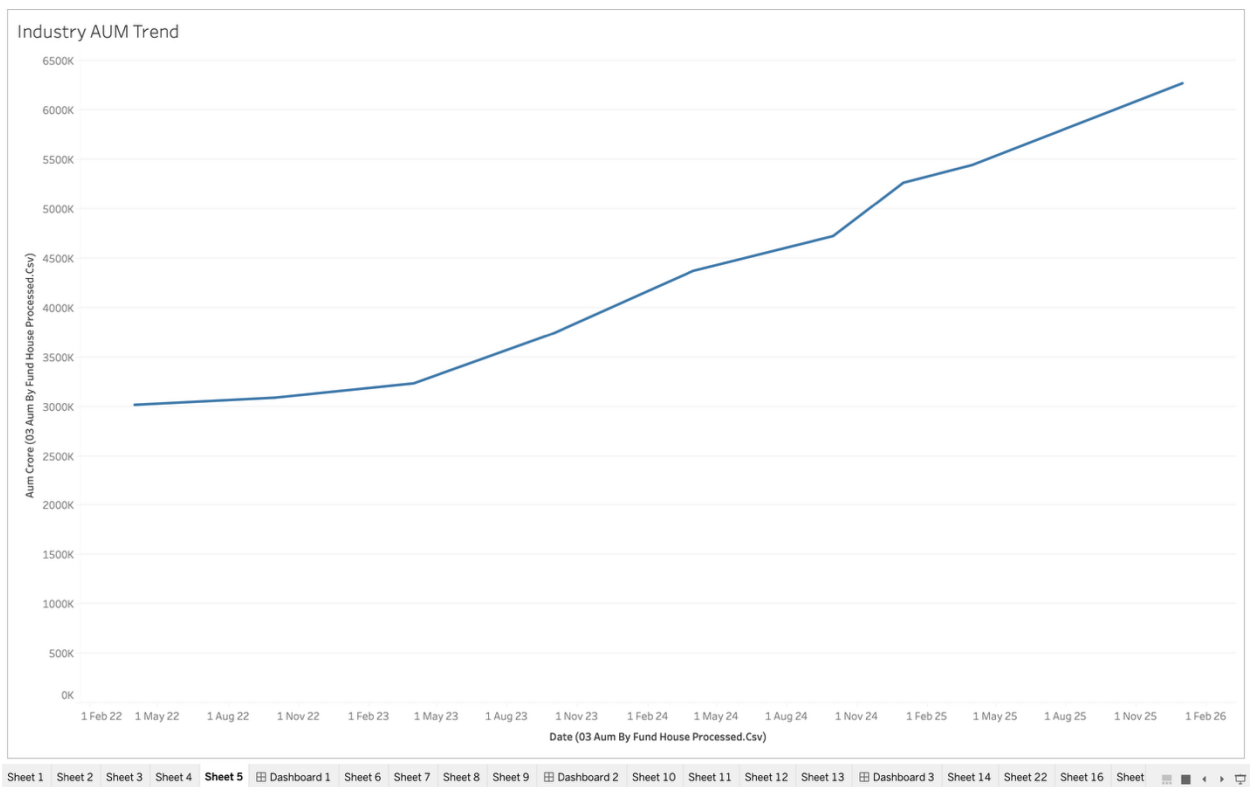


Fig 1 : Industry Growth Chart

SIP Growth Analysis

Monthly SIP inflows demonstrated strong growth trends.

Findings

- SIP contributions increased steadily year-over-year.
- Retail participation played a major role in market expansion.
- Consistent SIP growth indicates investor confidence in long-term investing.

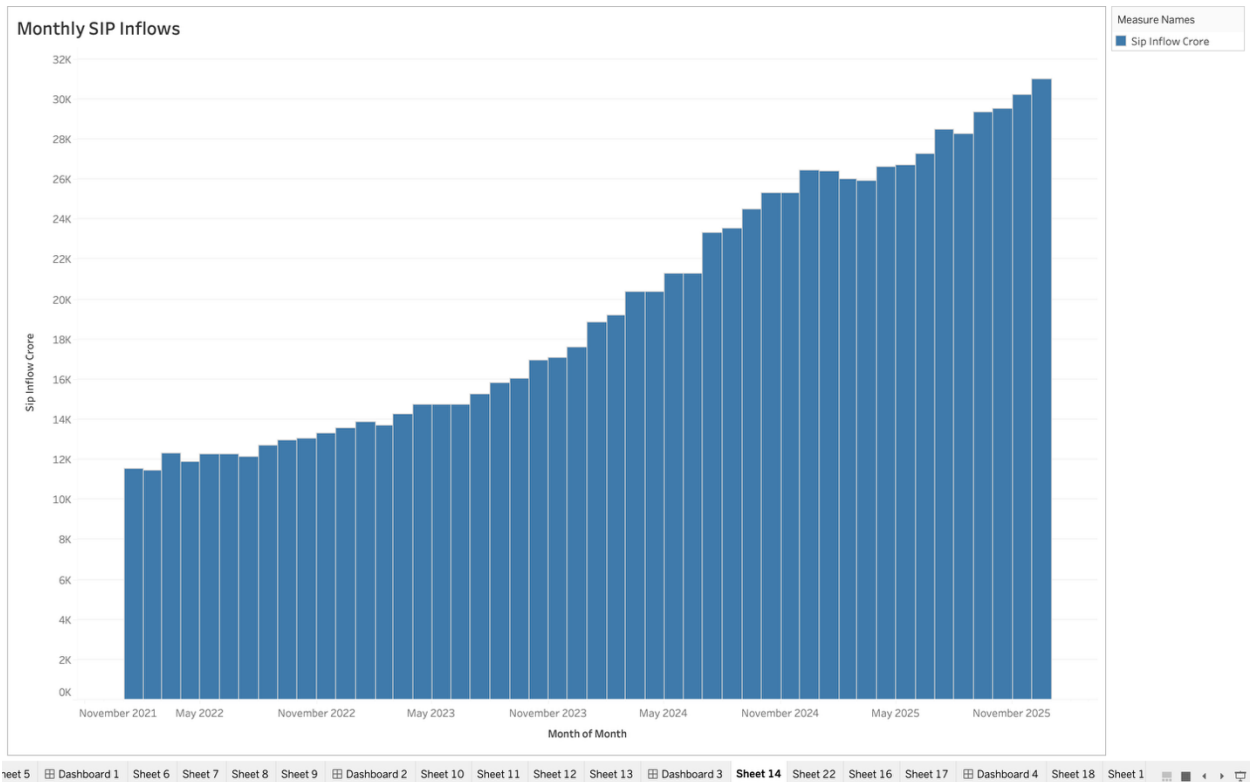


Fig 2 : Monthly SIP Inflows

Category Analysis

Category-level inflows highlighted investor preferences.

Findings

- Equity categories received the highest inflows.
- Small-cap and mid-cap funds experienced significant investor interest.
- Debt funds attracted conservative investors during volatile periods.



Fig 3 : Category Inflow Heatmap

7. PERFORMANCE ANALYSIS

Performance analysis focused on evaluating returns, risk, and risk-adjusted performance.

Return Analysis

Top-performing schemes generated significantly higher returns compared to category averages.

Observations

- Small-cap funds produced higher returns.
- Mid-cap funds offered strong growth opportunities.
- Large-cap funds provided stable returns.

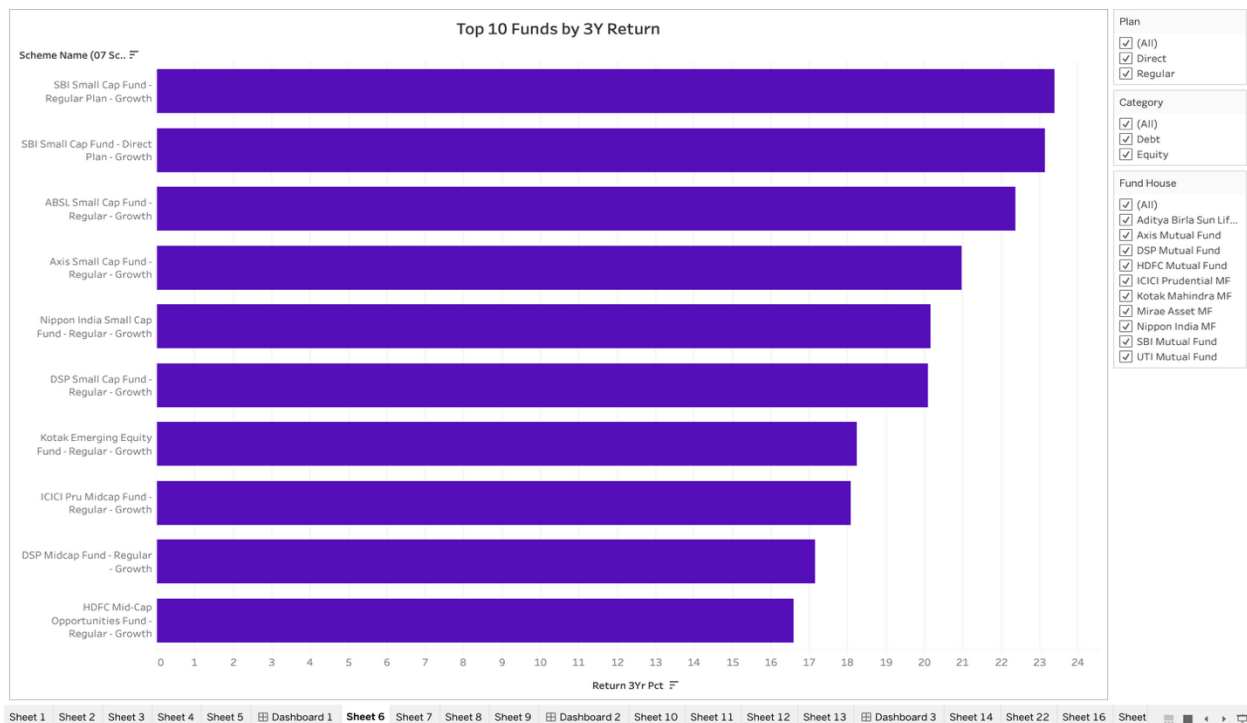


Fig 4: Top 10 Funds by 3Y Return

Risk Analysis

Risk was measured using annualized standard deviation.

Observations

- Higher-return funds generally exhibited higher risk.
- Risk-return trade-offs were clearly visible.

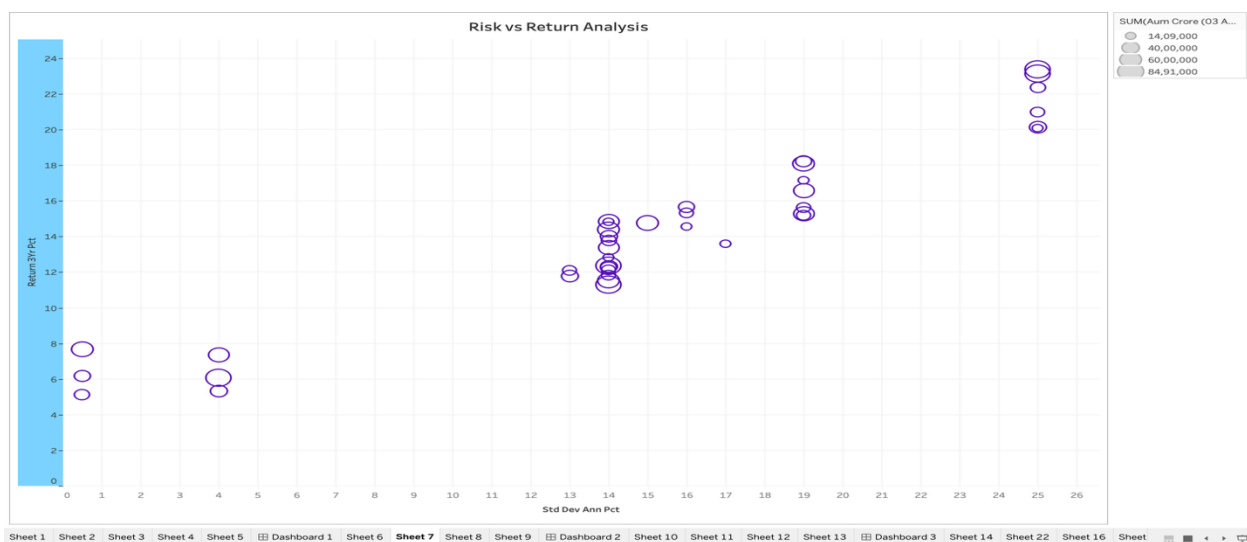


Fig 5 : Risk vs Return Analysis

Sharpe Ratio Analysis

Sharpe Ratio was used to evaluate risk-adjusted returns.

Findings

- Several funds achieved strong returns without excessive risk.
- High Sharpe Ratio funds offered superior investment efficiency.

8. ADVANCED ANALYTICS

Historical VaR (95%)

Historical Value at Risk estimates the maximum expected loss under normal market conditions at a specified confidence level.

Results revealed that small-cap funds exhibited higher VaR values, indicating greater downside risk.

Conditional VaR (CVaR)

CVaR measures average losses during extreme market conditions.

Findings showed that certain aggressive funds experienced larger losses during adverse periods.

Rolling Sharpe Ratio

Rolling Sharpe analysis revealed changing risk-adjusted performance over time.

Key Findings

- Market cycles significantly impacted Sharpe values.
- Some funds maintained consistent performance across periods.

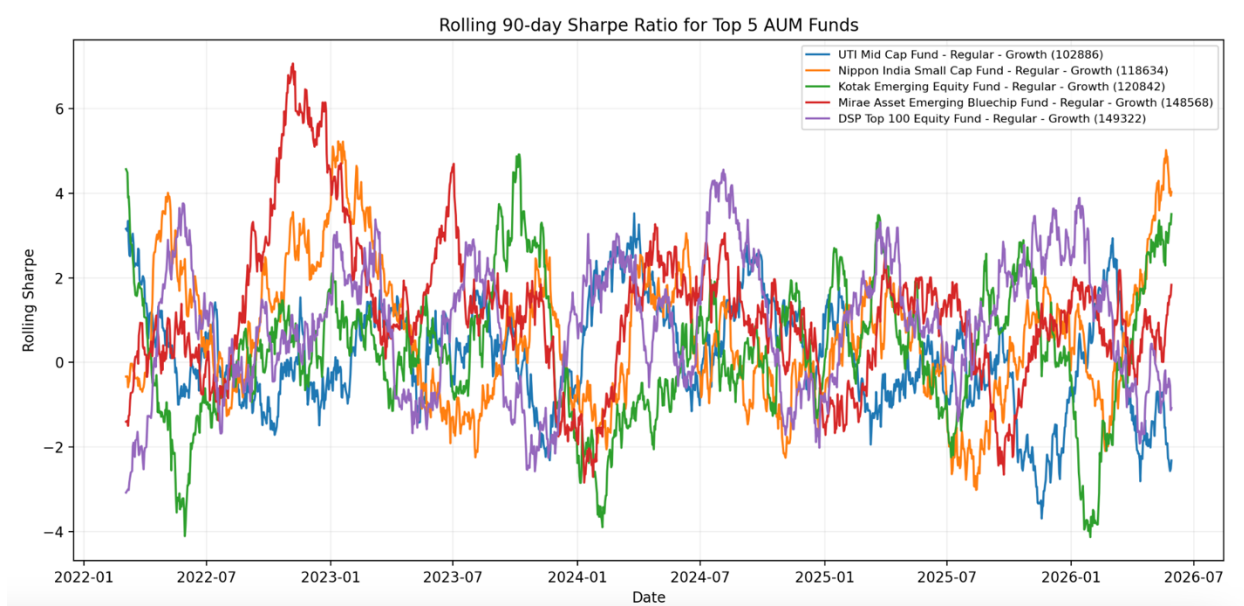


Fig 6 : Rolling Sharpe Ratio

Investor Cohort Analysis

Investors were grouped according to their first transaction year.

Findings:

- Early cohorts contributed larger cumulative investments.
- Newer cohorts displayed higher SIP participation rates.

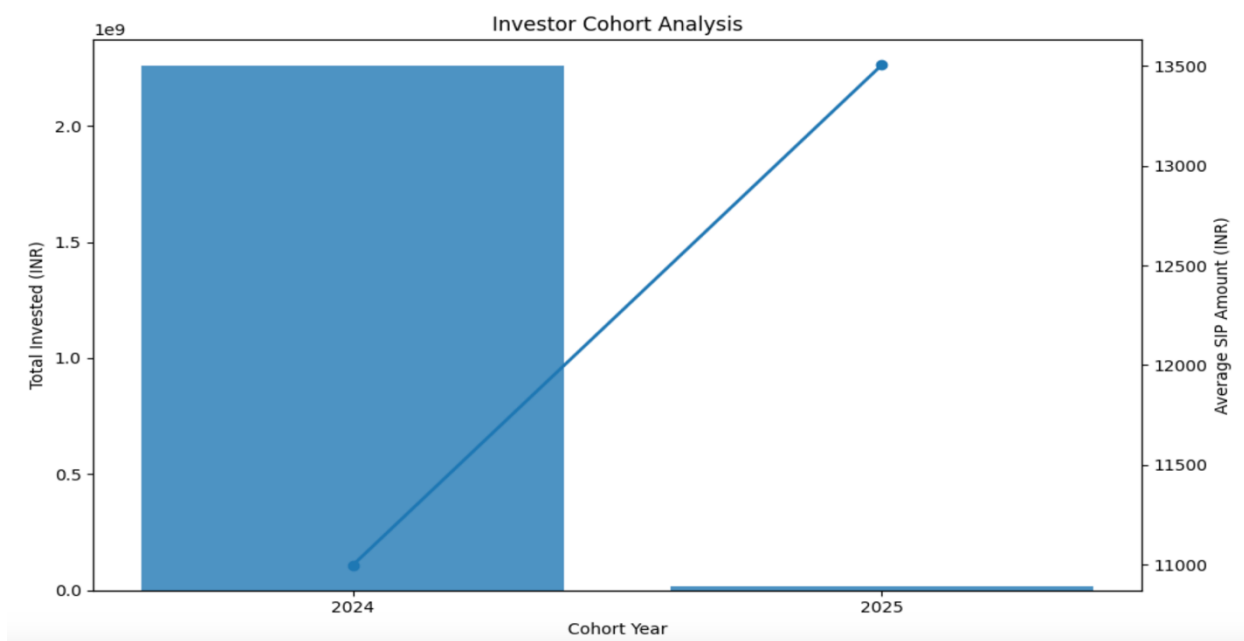


Fig 7 : Investor Cohort Analysis

SIP Continuity Analysis

Investor commitment was evaluated through transaction consistency.

Findings:

- Majority of investors maintained regular SIP schedules.
 - A subset of investors displayed increasing transaction gaps and were classified as at-risk.
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Sector Concentration Analysis

Herfindahl-Hirschman Index (HHI) was used to measure portfolio concentration.

Findings:

- Some equity funds maintained highly concentrated portfolios.
- Diversified funds demonstrated lower concentration scores.

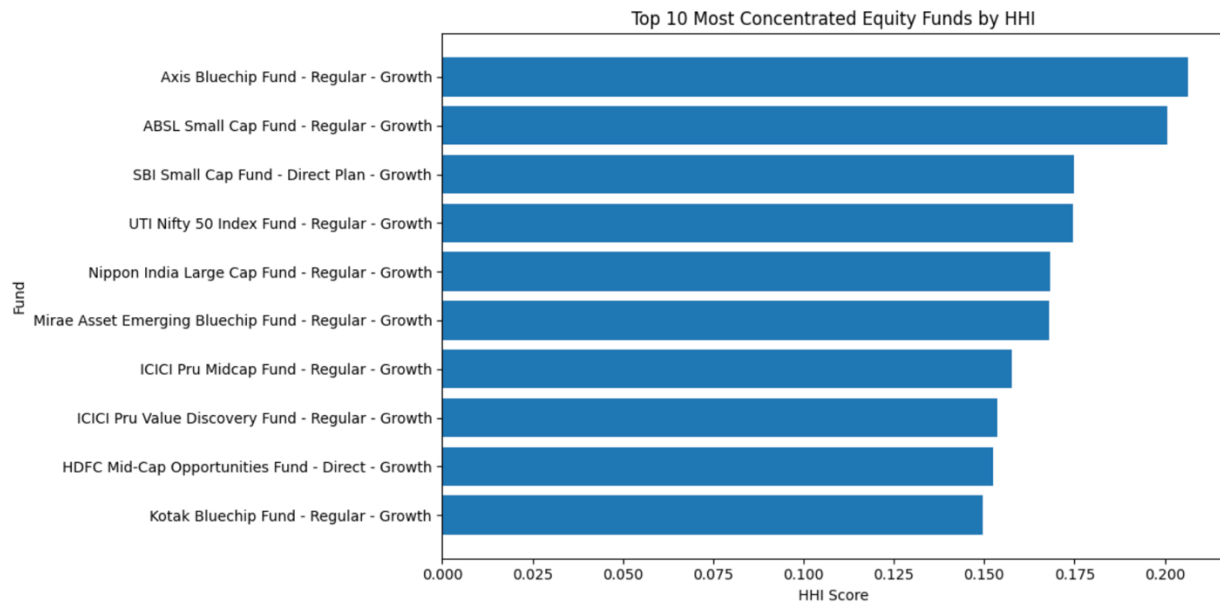


Fig 8 : Sector Concentration Analysis

9. DASHBOARD DEVELOPMENT

Interactive dashboards were developed using Tableau.

The dashboard suite contains four major analytical sections.

Dashboard 1 – Industry Overview

The Industry Overview Dashboard provides a high-level summary of the Indian mutual fund industry. The primary objective of this dashboard is to present key industry metrics and market growth trends in a concise and visually appealing format.

The dashboard includes KPI cards displaying:

- Total Assets Under Management (AUM)
- Total SIP Inflows
- Total Number of Schemes
- Total Investor Folios

These metrics provide a quick snapshot of the overall size and growth of the mutual fund industry.

A line chart was developed to visualize the industry AUM trend from 2022 to 2025. This chart helps identify long-term growth patterns and highlights the increasing scale of the mutual fund ecosystem.

An additional bar chart displays AUM by Asset Management Company (AMC), enabling comparison of market share across major fund houses. This visualization highlights industry leaders and reveals concentration within the asset management sector.

Dashboard Objectives:

- Monitor industry growth trends
- Track overall AUM expansion
- Compare major AMCs
- Present industry KPIs in a single view

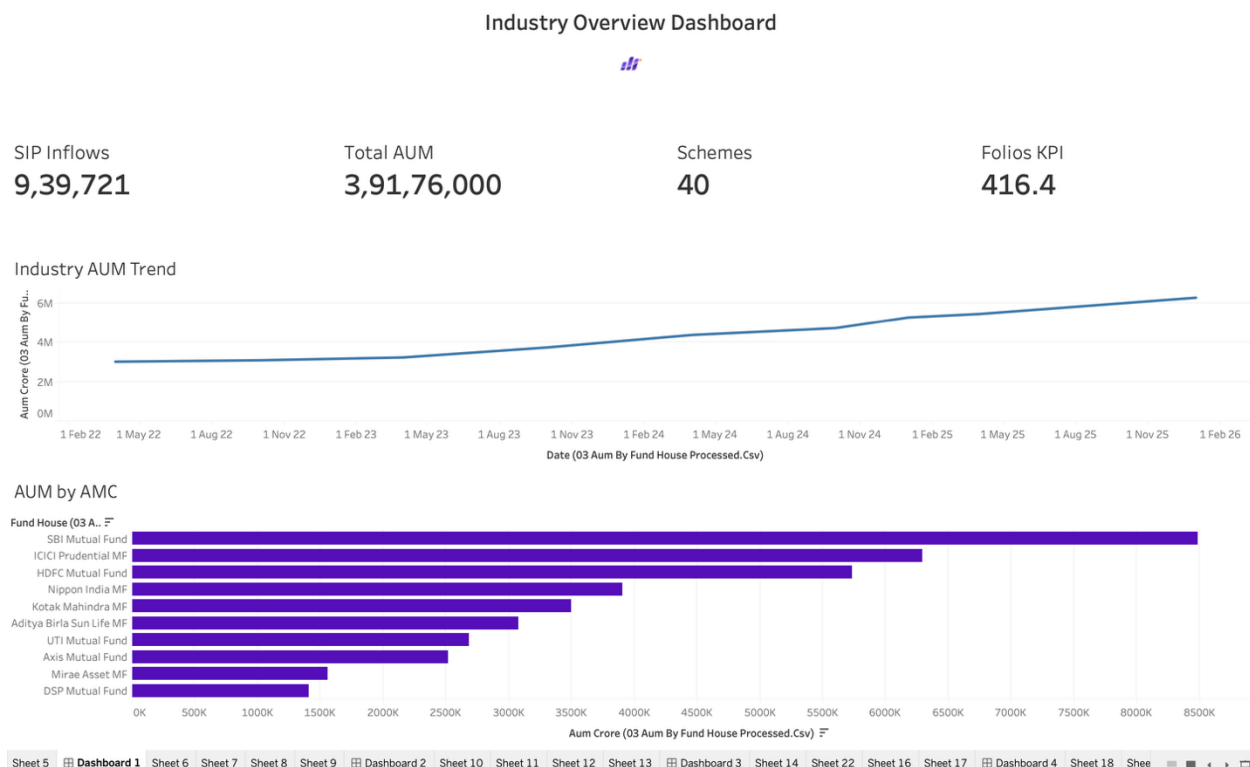


Fig 9 : Industry Overview Dashboard

Dashboard 2 – Fund Performance

The Scheme Performance Dashboard focuses on evaluating the performance and risk characteristics of individual mutual fund schemes.

A scatter plot was created to analyze the relationship between risk and return. Annualized standard deviation was used as the risk measure, while three-year returns were used as the performance measure. Bubble size represents fund AUM, allowing simultaneous comparison of risk, return, and fund size.

To identify top-performing schemes, ranking charts were developed for:

- Top 10 Funds by 3-Year Return
- Top 10 Funds by Sharpe Ratio

These charts help investors identify funds delivering strong absolute returns and superior risk-adjusted performance.

A category-level comparison chart was also developed to evaluate average returns across different mutual fund categories.

The dashboard includes a NAV vs Benchmark comparison chart, enabling users to assess whether fund performance is aligned with or outperforming benchmark indices over time.

Finally, a sortable fund scorecard table summarizes key metrics including:

- Fund Name
- AUM
- Standard Deviation
- Return Percentage
- Sharpe Ratio

Interactive filters allow users to analyze funds based on category, fund house, and investment plan.

Dashboard Objectives:

- Evaluate risk-adjusted performance
- Compare funds within categories
- Identify top-performing schemes
- Analyze benchmark tracking behavior

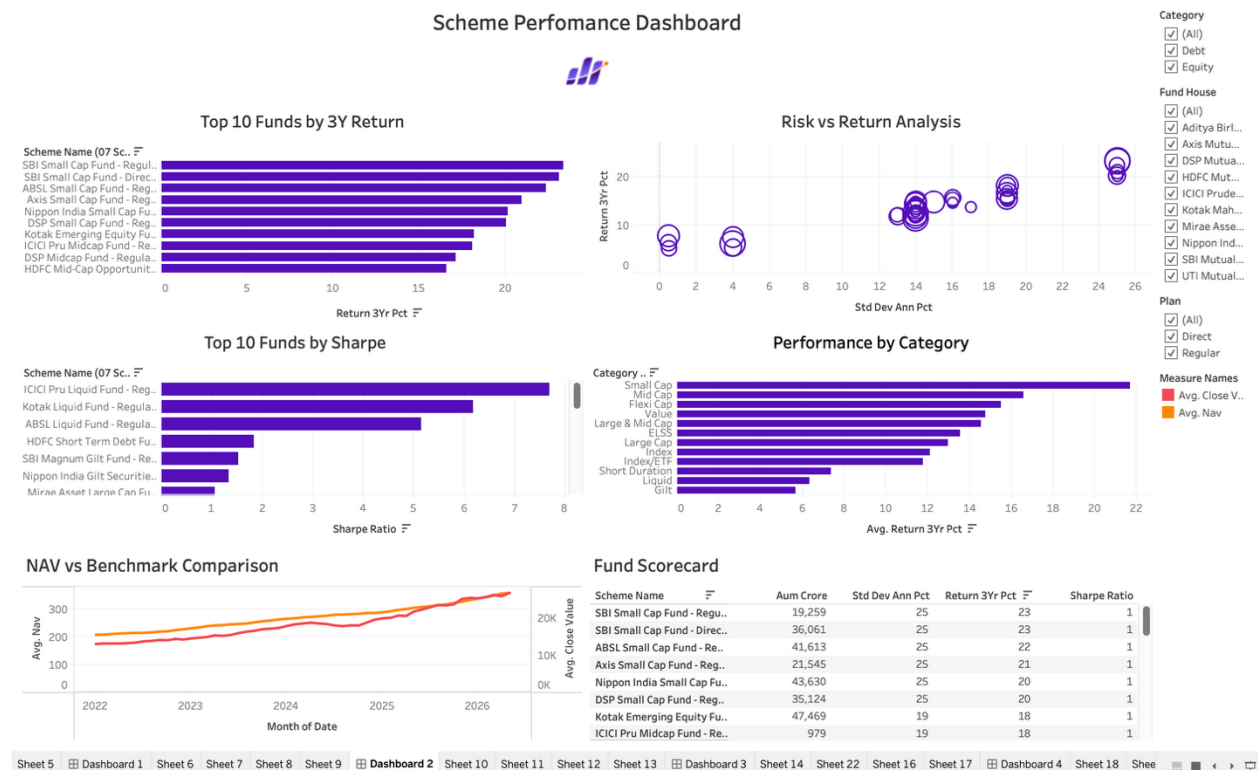


Fig 10 : Scheme Performance Dashbd

Dashboard 3 – Investor Analytics

The Investor Analytics Dashboard focuses on investor behavior, transaction patterns, and demographic insights.

A transaction amount by state visualization was created to identify geographical regions contributing the highest investment volume. This helps understand regional participation in mutual fund investments.

A transaction-type distribution chart categorizes transactions into:

- SIP Investments
- Lump Sum Investments
- Redemptions

This visualization provides insight into investor preferences and investment behavior.

An age-group analysis chart compares average investment amounts across different age segments. The analysis helps identify demographic groups contributing most actively to mutual fund investments.

A monthly transaction volume trend chart was developed to track investor activity over time and identify seasonal patterns in investment behavior.

Interactive filters allow exploration based on:

- State
- Age Group
- City Tier

These filters enable detailed demographic analysis and improve dashboard usability.

Dashboard Objectives:

- Analyze investor demographics
- Understand transaction behavior
- Identify geographic investment patterns
- Monitor investor participation trends

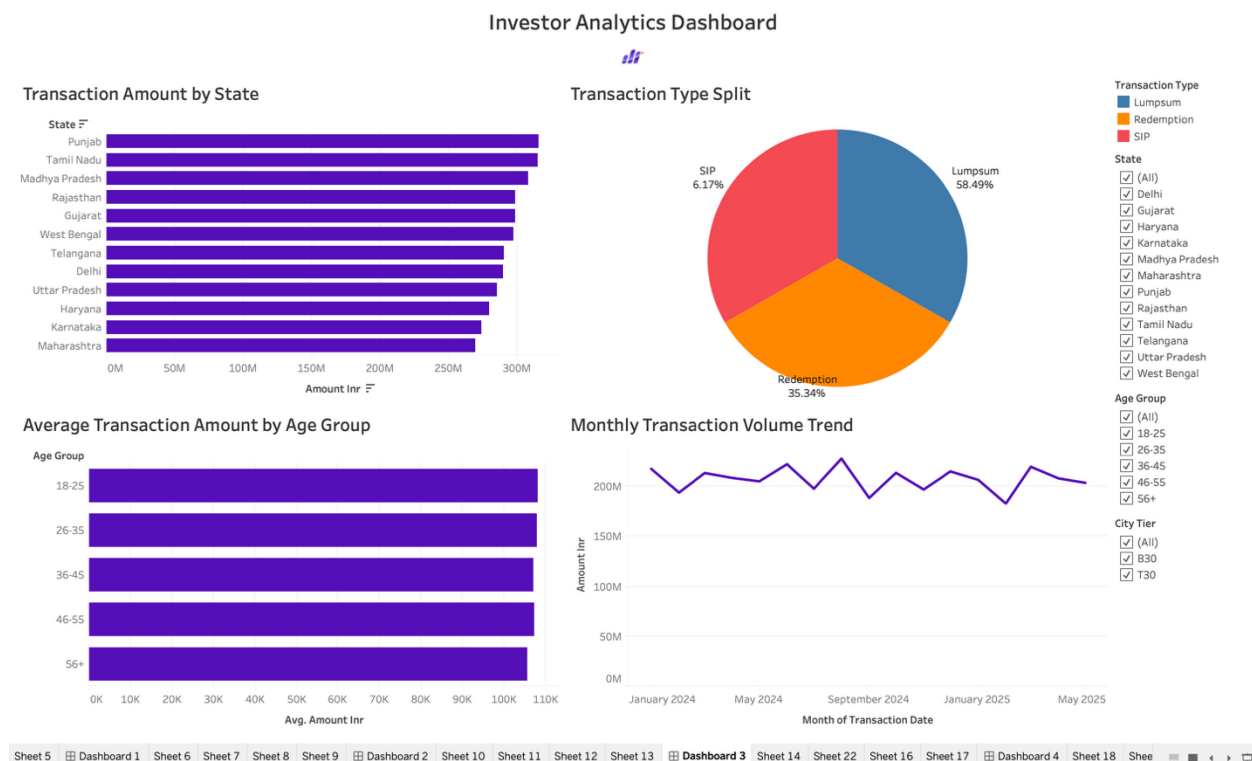


Fig 11 : Investor Analytics Dashboard

Dashboard 4 – SIP & Market Trends

The SIP and Market Trends Dashboard was designed to examine the relationship between investor participation and broader market movements.

A monthly SIP inflow visualization tracks the growth of systematic investment plans over time. The chart demonstrates increasing investor adoption of disciplined long-term investing.

A Nifty 50 trend chart was included to monitor benchmark market performance during the same period. This allows comparison between investor inflows and market movements.

A category inflow heatmap was developed to identify sectors and fund categories receiving the highest capital inflows. Darker cells indicate stronger investor interest within specific periods.

A Top 5 Categories by Net Inflow chart highlights categories attracting the highest investment capital, enabling identification of market trends and investor preferences.

Together, these visualizations provide insight into how investor sentiment and market performance influence capital allocation decisions.

Dashboard Objectives:

- Analyze SIP growth trends
- Monitor benchmark market performance
- Identify preferred investment categories
- Understand capital flow patterns

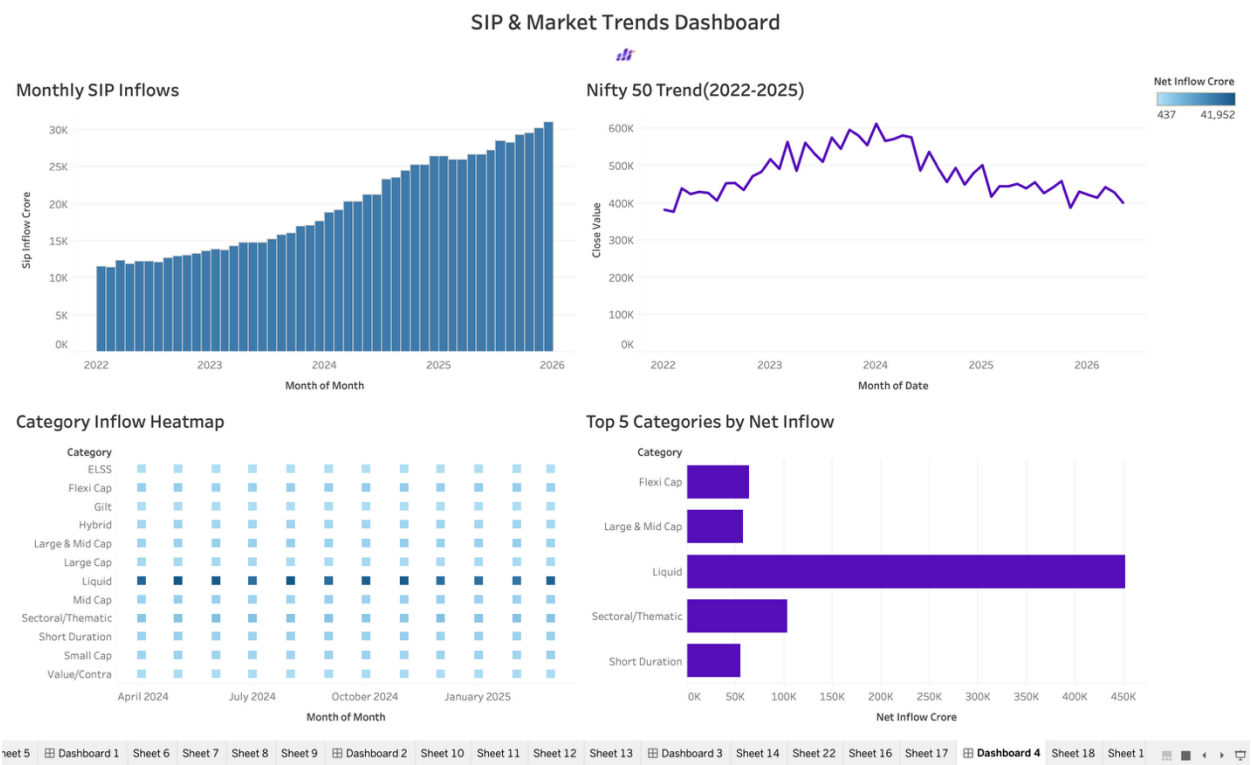


Fig 12 : SIP & Market Trends Dashboard

10. KEY INSIGHTS

1. Industry AUM exhibited sustained growth throughout the analysis period.
2. Small-cap funds generated the highest returns but also carried greater downside risk.
3. High-Sharpe funds delivered superior risk-adjusted performance.
4. SIP participation increased steadily, indicating growing investor confidence.

5. Investor cohorts entering earlier contributed significantly higher total investments.
 6. Certain investors demonstrated discontinuity in SIP behavior and were classified as at-risk.
 7. Portfolio concentration varied considerably across equity funds.
 8. Equity categories remained the preferred investment choice among retail investors.
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11. LIMITATIONS

Despite providing valuable insights, the project has several limitations.

- Dataset represents a limited historical period.
- Some data fields were synthetic or simplified.
- Risk calculations rely on historical behavior.
- External macroeconomic variables were not incorporated.
- Portfolio turnover analysis was not performed.

Future work may incorporate live market feeds and predictive modeling techniques.

12. RECOMMENDATIONS

Based on the analysis, the following recommendations are proposed:

1. Promote SIP-based investing due to demonstrated consistency and growth.
 2. Monitor investors identified as at-risk and implement retention strategies.
 3. Encourage portfolio diversification among concentrated equity funds.
 4. Utilize risk-adjusted metrics such as Sharpe Ratio for fund selection.
 5. Expand analytics capabilities with predictive forecasting models.
 6. Incorporate real-time market data integration.
 7. Develop personalized recommendation systems for investors.
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13. CONCLUSION

This project successfully developed a comprehensive Mutual Fund Analytics Platform that combines data engineering, financial analysis, and business intelligence to generate meaningful insights from mutual fund data. By integrating multiple datasets such as NAV history, AUM statistics, SIP inflows, investor transactions, portfolio holdings, and benchmark indices, the project created a unified analytical framework for evaluating fund performance and investor behavior.

Through the ETL process, raw data was cleaned, transformed, and prepared for analysis, ensuring consistency and reliability across all datasets. Exploratory data analysis revealed important trends such as the growth of industry AUM, increasing SIP participation, and investor preference toward equity-oriented schemes. Advanced analytical techniques including Sharpe Ratio, Value at Risk (VaR), Conditional Value at Risk (CVaR), investor cohort analysis, SIP continuity analysis, and portfolio concentration assessment provided deeper insights into risk, performance, and investment patterns.

The interactive Tableau dashboards further enhanced the usability of the analysis by presenting key findings through intuitive visualizations and filters. These dashboards enable stakeholders to monitor market trends, compare fund performance, and understand investor activity in a simple and effective manner.

Overall, the project demonstrates the practical application of data analytics and business intelligence in the financial domain. The developed platform provides a strong foundation for future enhancements such as predictive analytics, real-time market monitoring, and personalized fund recommendation systems, making it a scalable solution for data-driven decision-making in the mutual fund industry.

14. REFERENCES

1. (Association of Mutual Funds in India, n.d.) Available at: <https://www.amfiindia.com>
2. (National Stock Exchange of India, n.d.) NIFTY 50 Market Data. Available at: <https://www.nseindia.com>
3. Securities and Exchange Board of India (SEBI). Mutual Fund Regulations. Available at: <https://www.sebi.gov.in>
4. (Pandas Development Team, n.d.) Available at: <https://pandas.pydata.org/docs>
5. (Tableau Software, n.d.) Available at: <https://help.tableau.com>
6. Bluestock Fintech Capstone Project Datasets and Project Guidelines.